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Cheating in online courses: Evidence from online proctoring

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ABSTRACT

This study revives the unsettled debate on the extent of academic dishonesty in online courses. It takes advantage of a quasi experiment in which online proctoring using a webcam recording software was introduced for high-stakes exams in two online courses. Each course remained the same in its structure, content and assessments before and after the introduction of online proctoring. Analysis of exam scores shows that online proctoring was associated with a decrease in average performance in both courses. Furthermore, the decrease in scores persists when accounting for potential confounding factors in a regression framework. Finally, in separate regressions of exam performance on student characteristics, the regression explanatory power was higher for scores under proctoring. We interpret these results as evidence that cheating took place in the online courses prior to proctoring. The results also imply that online proctoring is an effective tool to mitigate academic dishonesty in online courses.

1. Introduction

In the past two decades, higher education institutions have experienced unprecedented growth in online learning. In the U.S., where this study took place, enrollment in distance higher education grew steadily between 2002 and 2016. Since 2012, whereas overall enrollment in higher education has been declining, growth in distance education has in fact been rising. As of 2016, the latest year for which data are published, close to a third of all college students were taking at least one distance course (Seaman et al., 2018).

More than half of these distance learners were students that were combining non-distance (face-to-face, F2F) learning with distance learning. Accordingly, today many "traditional" institutions offer a menu of online courses as well as fully online programs. This is prompted by sustained demand for such courses and programs – in 2016, for example, about 30 percent of students in public and private non-profit institutions in the U.S. enrolled in at least one distance learning course (*source*: own calculation using data in Seaman et al., 2018). It also appears that educators in all types of institutions have recognized that a structural shift has occurred, and that online delivery and learning will be a mainstay of higher education in the future.

Therefore, the dialogue surrounding online education has turned to how best to deliver online courses. Various aspects of online courses, such as modality (fully online versus hybrid; synchronous versus asynchronous), technology platform, assessment and accessibility are considered and debated. The goal of such dialogue, ultimately, is to design and deliver online courses in which student learning and experience are at least on par with traditional (F2F) courses. Given this goal, the question of how much learning takes place in online courses (relative to the traditional/F2F mode) has become a critical point of contention (see, among others, Cavanaugh & Jacquemin, 2015; Alpert et al., 2016; Dendir, 2019; Paul & Jefferson, 2019).

A particularly pertinent issue in this regard is academic dishonesty (McCabe et al., 2012). Some argue that even the measures that are used to gauge learning in online courses, such as scores on formative or summative assessments, do not truly reflect learning because they are possibly tainted by cheating that occurs during these assessments (Harmon et al., 2010; Arnold, 2016). If, for example, exam score distributions turn out to be comparable in an online course and its F2F counterpart, it does not mean that comparable learning takes place in the two modes simply because the online scores are likely inflated by cheating.[1] Such arguments are predicated on the assumption that academic dishonesty is more prevalent in online courses than F2F ones (Kennedy et al., 2000; Young, 2012).

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[1] There is a tangential discussion on whether high-stakes assessments measure learning regardless of the mode of delivery. For example, Wunder et al. (2013) argue that sometimes assessments may be gauging learning outcomes rather than learning.

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Various arguments are provided as to why online courses could be more amenable to academic dishonesty. One is that because assessments often happen in unsupervised or unproctored settings, it is difficult to confirm the identity of the test taker (Kraglund-Gauthier & Young, 2012). Similarly, online test takers can use unauthorized resources (e.g. cheat sheets, books or online materials) during assessments. Also, the online environment – by the mere absence of a close relationship and interaction with an instructor – can encourage collaborative (group) work with other students (Sendag et al., 2012; McGee, 2013; Hearn Moore et al., 2017).

While there is also growing empirical evidence showing that academic dishonesty is relatively more common in online learning, the debate is not yet fully settled (Harton et al., 2019; Peled et al., 2019). It is in this context that the current study presents evidence from a quasi/-natural experiment that occurred in two online courses at a midsize comprehensive university in the U.S. The experiment involved the introduction of online proctoring using a webcam recording software for high-stakes exams. The structure, content and assessments (exams) in each course remained the same before and after the introduction of online proctoring. A change in student performance, if any, can therefore be attributed to the mitigation of cheating after online proctoring came into place, and provides direct evidence on the scale of academic dishonesty in online courses.

Relative to much of the existing literature, the treatment here is unique because proctoring did not entail a change in modality. Many studies that investigate academic dishonesty typically compare student performance in *unproctored online* assessments and *proctored F2F* ones. But a comparison of student performance in proctored F2F exams and unproctored online exams may not be entirely valid because some of the performance differences could be due to the testing environment *per se*, apart from the effect of supervision (Fask et al., 2014; Butler-Henderson & Crawford, 2020). By comparing performance in the same learning mode (online) but before and after the advent of supervision, this study avoids any such complications.[2] Furthermore, from a practical point of view, in many scenarios in-person proctoring of tests may not be feasible for fully online courses. Therefore, the results of the current study also provide evidence on the efficacy of *easily adoptable, relatively low-cost* online proctoring in online courses.

The findings of the study suggest that cheating was taking place in the unsupervised exams. First, simple bivariate analyses show that there was a significant drop in average exam scores in both courses after online proctoring was introduced, in many cases by more than a letter grade. This is despite the fact that student characteristics remained largely similar before and after proctoring (implying sample selection was unlikely to be a factor). Second, explicitly accounting for student characteristics in a multiple regression framework could not explain away the decrease in performance. Finally, a comparison of the explanatory powers of regressions of scores on student ability and maturity indicators showed that they were higher for proctored exams. From these results one can also infer that online proctoring of assessments is a viable strategy to mitigate cheating in online courses.

The balance of the paper is organized as follows. The next section reviews the related literature on academic dishonesty. Section 3 describes the setup of the study and data. Section 4 presents bivariate analysis, the regression methodology and results. Section 5 points out some caveats and limitations of the study. The last section concludes and draws a few implications on the basis of the results of the study.

[2] In fact, Fask et al. (2014) suggest that some of the advantage that accrues from greater opportunities to cheat in an unproctored exam may be offset by the disadvantage of completing the exam in an online environment.

2. Literature

2.1. Why do students engage in academic dishonesty?

Academic dishonesty is any behavior intended to falsely represent one's academic work as original, and can take many forms, including deception, using unauthorized resources, collusion and plagiarism (Norris, 2019; Golden & Kohlbeck, 2020). A complex set of factors explains why students engage in academic dishonesty. In a basic model, McCabe et al. (2002) identify several direct reasons. These include, among others, lack of interest and/or preparation, pressures to perform, misconceptions about what constitutes academic dishonesty, lack of clarity in course and exam policy/expectations, a belief that others are engaged in similar behavior, and a perception, correctly or otherwise, that serious consequences are absent if one is caught cheating (see also Burrus et al., 2007; Jones, 2011; Yang et al., 2013). Which of these factors are at play depends on the type of academic dishonesty in question, particularly whether it happens spontaneously (known as panic cheating) or it is a result of forethought and deliberate preparation (known as planned cheating; Stone et al., 2014).

A more systematic view of the causes of academic dishonesty examines underlying motivations, contextual factors and dispositions of cheating students. Becker et al. (2006), adopting the fraud triangle framework used to diagnose sources of fraudulent behavior in business, model academic cheating as a function of three broad factors: *incentive* (motivation to cheat deriving from own/internal or external pressures), *opportunity* (ability to cheat because the environment allows it), and *rationalization* (ability to consider acts of cheating as not contravening with one's beliefs and ethics). In a related framework, Murdock and Anderman (2006) draw on motivational perspectives from various theories (self-efficacy, achievement goal, expectancy value and intrinsic motivation) to build an integrated model of academic dishonesty.

A slightly different but complementary strand of the literature augments these models by exploring the role of personality traits in cheating. In particular, studies have theorized how the Big Five personality factors (neuroticism, extraversion, openness to experience, agreeableness and conscientiousness) can increase or attenuate academic dishonesty (Nathanson et al., 2006; Williams et al., 2010). More recently, the Theory of Planned Behavior has been employed to understand and predict academic dishonesty (Mayhew et al., 2009; Cronan et al., 2015; Chudzicka-Czupala & Grabowski, 2016; Lonsdale, 2017). In this model, the intention to cheat and cheating behavior become a function of *own beliefs* (related to anticipated outcomes), *normative beliefs* (related to reference individuals or groups) and *control beliefs* (related to the challenge or obstacles to an action).

Largely building on these frameworks, several studies have empirically investigated the causes of academic dishonesty. Becker et al. (2006) find evidence that all three elements of the fraud triangle drive cheating behavior among business students. Heavy workload and high expectations from friends and family are shown to cause students to commit academic dishonesty (Finchilescu & Cooper, 2018; Jian, Li, & Wang, 2020). In contrast, pedagogical approaches that prize learning and deemphasize the role of grades diminish incentives for and occurrence of cheating (Day et al., 2011). Pulfrey et al. (2019) report that in classrooms, inducing a mastery-approach (in lieu of a performance-approach) and autonomy (in lieu of control) reduces academic dishonesty.

Environments that do not have clear rules on academic integrity, where expectations and sanctions are not well-stipulated, and enforcement and penalties may be lacking foster academic dishonesty (Finchilescu & Cooper, 2018; Peled et al., 2019; Englander et al., 2011). In this regard, a large body of evidence has documented the important role that honor code systems, and faculty and institutional efforts to raise students' knowledge of academic integrity rules, play in reducing the perception and incidences of academic misconduct (see, among others, McCabe et al., 2002; Arnold et al., 2007; Burrus et al., 2007; Tatum & Schwartz, 2017).

Psychological dispositions also seem to make some students more likely to engage in academic dishonesty than others. Although the evidence is mixed, impulsivity (an expression of neuroticism) and a high need for sensation (an expression of extraversion) are often thought to be positively associated with accepting attitudes toward cheating and tendencies to cheat (Wilks et al., 2016; Anderman & Won, 2018; Peled et al., 2019). In contrast, students that rank high in traits related to the construct of conscientiousness (e.g. prudence, self-control and purpose) are more likely to consider cheating as unacceptable and less likely to commit acts of academic misconduct (Kisamore et al., 2007; Williams et al., 2010; Day et al., 2011; Wilks et al., 2016; Yu et al., 2017; Peled et al., 2019). However, a recent meta-analytic study concluded that of the Big Five factors, the two strongest predictors are agreeableness and conscientiousness, with both of them negatively related to academic dishonesty (Giluk & Postlethwaite, 2015).

More broadly, social norms and culture are also found to play a role in academic dishonesty in higher education. Using an expanded version of the Theory of Planned Behavior that adds *moral obligation* as a fourth factor, Chudzicka-Czupala et al. (2016) analyze intentions to cheat among university students in seven countries. They find that moral obligation, which is related to feelings of guilt and shame, indeed is a strong predictor of academic dishonesty in all samples (see also Orosz et al., 2013). However, whereas own beliefs also feature prominently in individualistic societies like the United States and Switzerland, social/-subjective norms are more important in affecting cheating behavior in collectivist cultures like the Ukraine. Maloshonok and Shmeleva (2019) find a similar result in Russian universities, whereby subjective norms – specifically, undergraduates' perception of what their peers do – supersede own beliefs in determining involvement in academic dishonesty.

Finally, several studies have examined potential linkages between academic misconduct and students' demographics, characteristics and behaviors, although findings have been mixed and consistent patterns have been hard to identify. The bulk of the evidence seems to suggest that younger students (e.g. freshmen) and males are more likely to cheat (Schuhmann et al., 2013; Hensley et al., 2013; Wray et al., 2016; Yu et al., 2017; Lento et al., 2018). However, some studies report negligible or opposite effects of age and gender (Nathanson et al., 2006; Yang et al., 2013; Owens, 2015). Greek society (fraternity/sorority) membership, poor academic performance and alcohol consumption are associated with higher chances of cheating (Burrus et al., 2007; Rigby et al., 2015). Academic dishonesty appears more commonplace in some majors (e.g. engineering and business; see (Carpenter et al., 2006; Simha et al., 2012), although this finding has been contested as well (Iyer and Eastman, 2006; Yang et al., 2013). Finally, prior experience with academic dishonesty (e.g. in high school), risk-taking behavior and personally focused values (e.g. hedonism, power) are shown to increase the probability of cheating in college (Schuhmann et al., 2013; Rigby et al., 2015; Xu et al., 2019; Koscielniak & Bojanowska, 2019).

2.2. Academic dishonesty in F2F vs. online and proctored vs. unproctored environments

The research evidence on the prevalence of academic dishonesty comes from two types of studies – *survey-based* and *performance-based* studies. Survey-based studies gather information on attitudes, perceptions and behavior related to academic dishonesty by conducting surveys of students and faculty. Therefore, the evidence presented by these types of studies is principally self-reported.

When it comes to perceptions or beliefs of students, the early evidence from survey-based studies painted a decidedly mixed picture on whether academic dishonesty is more prevalent in F2F or online courses. Some studies reported that students perceive levels of academic dishonesty to be comparable in traditional and online courses (Spaulding, 2009; Harmon et al., 2010). Other studies indicated that, in line with generally held beliefs, students perceive cheating to be more commonplace in online courses (King et al., 2009; Watson & Sottile, 2010). Still others presented contrary evidence, showing that students believe more cheating happens in the traditional/F2F environment than online (Black et al., 2008). Interestingly, Harmon et al. (2010) noted that, regardless of course format, students believe proctoring is the main factor that mediates cheating, especially when assessment is mostly done using multiple-choice questions.

Whereas the aforementioned studies gauged students' beliefs and perceptions, much of the early evidence on self-reported acts of academic dishonesty – that is, self-reported *behavior* – was also mixed. Some studies reported similar levels of academic dishonesty between live (F2F) and online classes (Grijalva et al., 2006; Spaulding, 2009). Other studies documented that self-reported cheating was generally more prevalent in online classes (Lanier, 2006; Miller & Young-Jones, 2012). Finally, there are also studies which reported higher levels of self-reported academic dishonesty in traditional courses (Kidwell & Kent, 2008; Watson & Sottile, 2010; Stuber-McEwen et al., 2009; Sendag et al., 2012).

How can there be more academic dishonesty in traditional courses? One possible explanation is that the close social interaction that develops among students in traditional classes can foster a climate of peer pressures as well as collusion. Another possible explanation is that most online-only students tend to be older, more mature and motivated, and thus generally less likely to engage in cheating (Kidwell and Kent, 2008). For instance, lower levels of cheating are reported by students enrolled only in online courses, whereas those taking both online and traditional courses think more cheating occurs in the online environment (Miller & Young-Jones, 2012). Similarly, Sendag et al. (2012) noted that, compared to students that took at least one hybrid or online course, those entirely enrolled in traditional courses were more likely to engage in online academic dishonesty (e-dishonesty).

More recently, however, survey-based studies are increasingly reporting that academic dishonesty – in perception as well as self-reported behavior – is more common in online environments. In surveys, students convey that more cheating occurs in online exams, typically by way of looking up answers on various devices, using social media, or by seeking unauthorized help from friends (e.g. Owens, 2015; Best & Shelley, 2018). The finding cuts across disciplines. Engineering students report higher tendencies to cheat, a phenomenon that may worsen in online settings (Srikanth & Asmatulu, 2014). Both undergraduate and graduate business students also perceive cheating to be more common in the online environment (Larkin and Mintu-Wimsatt, 2015). According to Harton et al. (2019), a majority of students (and faculty) from several disciplines surveyed at a midsize public university think cheating and plagiarism are bigger problems in online than F2F classes. A clear exception to these group of studies is Peled et al. (2019) who, using data from six institutions in the United States and Israel, report that surveyed students were less likely to participate in academic dishonesty in online courses than F2F courses.

Again, survey responses make clear that proctoring is critical. Owens (2015) found that frequency of engaging in cheating behavior was higher in unproctored online tests, while there were no differences in reported cheating between F2F and remotely proctored online tests. Dyer et al. (2020) investigated perceptions and behavior related to proctored and unproctored exams by surveying first- and second-year undergraduate engineering students in four universities. The results were rather critical of unproctored exams – students reported both that they felt cheating was more acceptable and that they were more likely to engage in a variety of cheating behaviors in an unproctored (typically online) environment. The authors conclude that students in distance education consider proctoring as a signal that an institution takes assessments and, more generally, academic integrity seriously.

The drawback of survey-based studies is that they may lack accuracy and validity because academic dishonesty is a sensitive issue and survey participants may not be entirely forthcoming in their responses (Howard, 2019). Even when truthful though, perceived or anticipated behavior can often deviate from actual behavior, and students also often overestimate the degree to which others engage in dishonest behavior (Hard et al.,

2006). Therefore, the second group of studies attempts to measure the prevalence and extent of academic dishonesty by analyzing performance data (hence named *performance-based* studies). These studies often seek to find direct evidence of cheating by comparing student performance in supervised (typically F2F) and unsupervised (typically online) environments.

However, the accumulated evidence from performance-based studies is not entirely definitive either. On the one hand, there are studies which simply failed to find an association between proctoring and lower average performance, and consequently suggested that academic dishonesty is not a big issue in unproctored environments (Yates & Beaudrie, 2009; Hollister & Berenson, 2009; Golden & Kohlbeck, 2020; Ladyshewsky, 2015; Rios & Liu, 2017). Beck (2014), adopting a methodology pioneered by Harmon and Lambrinos (2008), compared the explanatory power of a regression of proctored exam scores on student characteristics to one from a regression of unproctored exam scores. But the explanatory power of the regression model was not found to be consistently higher in monitored exams, again leading to the inference that cheating did not play a significant role in unmonitored settings.[3]

On the other hand, a series of studies are increasingly documenting that academic dishonesty is indeed a significant problem in unproctored environments. Some studies present direct evidence, whereby mean student performance is significantly lower (sometimes by 10–20 percentage points) when exams are administered with proctoring than without (Fask et al., 2014; Alessio et al., 2017; Daffin & Jones, 2018). Earlier, in a pioneering study, Harmon and Lambrinos (2008) had compared the explanatory power of a regression of proctored exam scores to that of unproctored exam scores. The explanatory power was much higher for the proctored exam, a result they interpreted as evidence that cheating, an unobserved behavior, skewed the scores in the unproctored exam. Adopting a similar methodology, Arnold (2016) examined whether cheating occurs in unproctored formative online tests in several courses given in a first-year curriculum at a large school of economics. The relationship between student characteristics and exam performance was tighter in proctored formative tests than unproctored ones, which suggests cheating taking place in the latter (although statistically the results were not as strong as Harmon & Lambrinos, 2008).[4]

Even in instances when there is no clear evidence of poorer performance under proctoring, other metrics raise suspicion about unproctored environments. Hylton et al. (2016) compared online exam performances of two groups of students, where one group was monitored by a web-based proctor while the other group was unmonitored. The results showed that although differences in performance were not statistically significant, the unmonitored group took longer time during tests. Howard (2019) found a similar pattern when comparing test scores of three groups of students: online (unproctored), onsite (proctored), and online proctored (using remote software). Again mean performance of the unproctored group was comparable to that of the two proctored groups. But unproctored students generally spent more time on exams, which suggests they were attempting to search for answers during tests.

In a recent insightful study, Golden and Kohlbeck (2020) compared performances of online students that were given verbatim test questions to those given paraphrased questions. Those who did the test bank questions scored higher (by 11 percentage points), which suggests students were looking up solutions for test bank questions on the internet. Interestingly, the gap in performance was smaller when the tests were done under proctoring technology, implying that proctoring partially alleviates cheating.

3. Study setting and data

3.1. Setting

The current study took place in two online courses that were offered at a comprehensive midsize public university in the U.S. The courses were principles of microeconomics (offered by the Department of Economics) and geography of North America (offered by the Department of Geospatial Science). Both were well-established, fully asynchronous online courses taught by two experienced instructors. They were delivered primarily through the online learning management system known as Brightspace (by Desire-to-Learn, D2L).

Specifically, the current study is based on nine sections of principles of microeconomics (offered between Spring 2015 and Spring 2019) and ten sections of geography of North America (offered between Fall 2014 and Spring 2019). Each course remained virtually unchanged in its structure, contents and assessments (exams) during the relevant study period. However, in Spring 2018 a major development occurred that significantly changed how high-stakes assessments were done in the two courses. This involved the adoption of online proctoring through a webcam recording software known as Respondus Monitor (RM).

In each course there were three exams that in combination accounted for 45 to 70 percent of the final course grade. Each exam roughly covered a third of the course and was noncumulative. Exams were predominantly made up of multiple-choice questions. Before Spring 2018, exams were completed online in the absence of any direct supervision (or proctoring), although several steps and techniques were adopted to mitigate cheating. These included the use of a special browser (Respondus Lockdown Browser) to prevent other applications from opening during a test, automated randomization of order of test questions and choices, limited time window (period) in which students could attempt a test, and a strict limit (timer) on test duration.

The use of RM starting in spring 2018 introduced direct proctoring, albeit in the same online environment. When using RM, a student is first guided through a series of pre-exam procedures in which he or she is given exam instructions and expectations. As part of these steps, the student is also photographed and video recorded while showing proof of identification and the surrounding environment. Subsequently, RM will record the student for the entire duration of the exam. Each student's recording will be made available to the instructor shortly after the conclusion of the exam, along with still pictures taken of the student at random times during the test. The software also includes a report with flags on any irregular incidents (e.g. reduced visibility or face recognition, or detection of an unrecognized person in the camera frame) that may have occurred and their exact timeline.

Two types of data are used for the analysis in this paper. Exam scores were accessed from instructor records. These were then matched with student demographic and academic background variables that were obtained from the university's institutional research office.[5] The purpose of collecting information on these additional variables is to check if there was any systematic change in student composition that somewhat coincided with the treatment (introduction of online proctoring) and also to use as explanatory variables in multiple regressions of student performance.

The literature has identified a student's human capital (or ability), gender and maturity to be potential determinants of performance in a college course. Following guidance from the literature, a student's grade

[3] In a comment, Englander et al. (2011) had provided a critique of Yates and Beaudrie (2009), pointing out potential deficiencies in the setup and analysis of the study that could have led to the absence of a significant difference in grades from monitored and unmonitored exams. In part, Beck (2014) was trying to provide fresh evidence on the issue while addressing many aspects of the Englander et al. (2011) critique.

[4] Interestingly, the study also finds that cheating does not pay off. Students who achieved higher scores on unproctored formative tests did not necessarily perform well on proctored summative exams. Students that were statistically more likely to have cheated on unproctored tests were also found to have an overall lower probability of succeeding in college (Arnold, 2016).

[5] The study was approved by the university's Institutional Review Board (IRB).

Table 1a
Summary statistics on student characteristics, Economics.

	Unproctored			Proctored			t	P > t
	Mean	Std. dev	N	Mean	Std. dev.	N		
Female	0.38	0.49	133	0.32	0.47	75	0.81	0.42
Age	20.29	2.12	133	19.99	1.43	75	1.09	0.28
GPA	2.85	0.63	133	2.65	0.66	75	2.21	0.03
Credit hours	50.53	30.59	133	46.75	26.77	75	0.90	0.37

Table 1b
Summary statistics on student characteristics, Geography.

	Unproctored			Proctored			t	P > t
	Mean	Std. dev	N	Mean	Std. dev.	N		
Female	0.52	0.50	288	0.55	0.50	152	0.63	0.53
Age	21.40	3.13	288	20.77	3.75	152	1.87	0.06
GPA	2.88	0.63	288	2.89	0.61	152	0.15	0.88
Credit hours	76.77	33.97	288	68.34	34.62	152	2.46	0.01

point average (GPA) is used to proxy ability (Grove et al., 2006; Arnold, 2016). Specifically, a student's incoming GPA is used, which is the GPA before the beginning of the semester in which the course was taken. If incoming GPA is not available, which may be the case if a student is a new freshman or newly transferred to the college, end-of-semester GPA was used instead, but one that excludes the grade in the course in question. Age is used to proxy a student's maturity. Maturity is not only a possible determinant of course performance but it has also been associated with a lower likelihood that a student engages in acts of academic dishonesty. In addition to age, cumulative credit hours at the beginning of the course was used to measure a student's college-specific experience.[6]

3.2. Data

Tables 1a and 1b presents descriptive statistics on the study sample. The main criterion used to construct the sample was that a student must have stayed the duration of the course (i.e. must have taken all three exams). After applying this criterion and dropping observations with missing information on other important variables, the sample size was 648 students (208 in economics and 440 in geography). The statistics are summarized by course and proctoring status – unproctored and proctored (that is, before RM/spring 2018 versus after RM/spring 2018). A test of equality of means is reported for each variable to check if there has been a significant change in sample composition following the introduction of online proctoring.

Table 1a shows that in the economics course, the sample under online proctoring had slightly fewer females and was, on average, younger and less experienced (in starting credit hours) than the sample before online proctoring. However, none of these differences were statistically significant. But the average GPA was lower under online proctoring, a difference that was statistically different from zero ($t = 2.21$, $p\text{-value} = 0.03$).

In the geography course (see Table 1b), the proportion of female students and average GPA were statistically indistinguishable between the samples with and without proctoring. However, the post-RM sample in geography was younger ($t = 1.87$, $p\text{-value} = 0.06$) and had, on average, eight fewer accumulated credits at the beginning of the course ($t = 2.46$, $p\text{-value} = 0.01$).

One of the aims of the current study is to seek evidence on academic dishonesty in online learning that is robust across academic disciplines – hence the use of data from two different courses. In a majority of the performance-based studies reviewed above, the evidence was limited to one particular field of study (or school/department) or cohort. The two courses in this study come from different academic colleges in the university (College of Business and Economics and College of Science and Technology). Furthermore, although both courses are designated as 200-(sophomore) level, as Tables 1a and 1b reveal there are notable differences in the characteristics of students taking them. Specifically, relative to the economics course, a typical student in the geography course was more likely to be a female, older, more experienced (a junior versus a sophomore) and of higher ability.

[6] Although it must be noted that for the traditional college age group (18–22 years), whether cheating behavior decreases with age (or its close correlate, class rank) is less clear (McCabe et al., 2012).

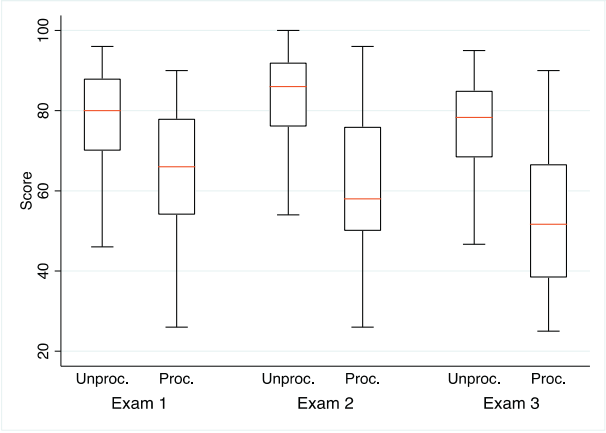


Fig. 1a. Distribution of exam scores by proctoring status, Economics.

4. Results

4.1. Bivariate analysis

The simplest and most direct way to assess whether cheating was taking place in these two courses in the absence of proctoring – and subsequently determining whether online proctoring is effective – is by comparing descriptive statistics of student performance before and after the introduction of RM. Fig. 1a and b presents, respectively for economics and geography, boxplots of student performance by proctoring status in the three exams.

It is clear from the boxplots that proctoring has entailed a downward shift of the performance distribution in every exam given in the two courses. The median score is substantially lower under proctoring, in some cases by upwards of 20 percentage points. The dispersion (variability) in scores is also generally larger for proctored exams, mainly precipitated by the much lower scores in the bottom part of the score distribution.

Similarly, Tables 2a–2b presents summary statistics (means and standard deviations) on performance in the three exams given in economics and geography, respectively. These statistics are again accompanied by tests of equality of mean scores across the two regimes. Here as well, it is clear that in both courses average performance was much lower when proctoring was in place. The difference in corresponding mean exam scores before and after proctoring ranged from 10 to 20 percentage points, and was generally larger in economics (the average difference was 18.6 percentage points in economics and 13.5 percentage points in geography). All the differences were also highly statistically significant.

In combination, the overall changes in the score distributions and the considerable drop in average performance strongly suggest that academic dishonesty was likely a common occurrence prior to the introduction of online proctoring. They also imply that online proctoring worked effectively as a tool to mitigate cheating during high-stakes

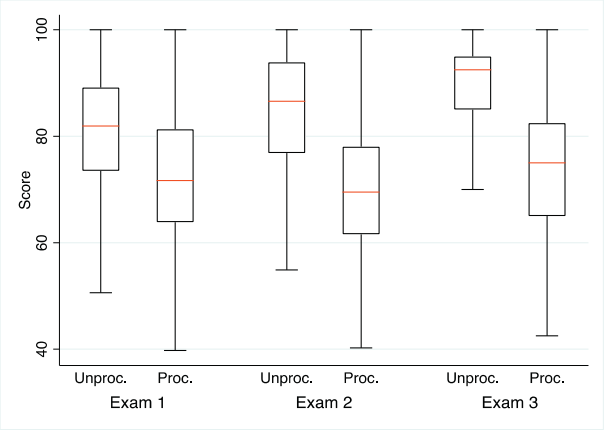


Fig. 1b. Distribution of exam scores by proctoring status, Geography.

Table 2a
Summary statistics on exam scores, Economics.

	Unproctored			Proctored			t	P > t
	Mean	Std. dev	N	Mean	Std. dev.	N		
Exam 1	78.39	12.04	133	64.93	14.40	75	7.20	0.00
Exam 2	82.77	12.34	133	62.27	16.79	75	10.07	0.00
Exam 3	74.67	14.49	133	52.80	16.81	75	9.86	0.00

Note: Exam scores are in percentage.

Table 2b
Summary statistics on exam scores, Geography.

	Unproctored			Proctored			t	P > t
	Mean	Std. dev	N	Mean	Std. dev.	N		
Exam 1	81.47	10.67	288	72.00	12.86	152	8.24	0.00
Exam 2	84.62	11.15	288	69.90	13.23	152	12.33	0.00
Exam 3	88.85	9.83	288	73.59	13.40	152	13.61	0.00

Note: Exam scores are in percentage.

exams. Whereas these ANOVA results are strongly suggestive, the next subsection discusses the multiple regression-based analysis that will be used to further examine academic dishonesty in these courses.

4.2. Regression analysis

4.2.1. Methodology

In this paper, the regression methodology of Harmon and Lambrinos (2008) is adopted to assess academic dishonesty among online students. As noted earlier, the methodology was also used by Beck (2014) and Arnold (2016). The model regresses test scores on measures of human capital (or ability) and other potentially relevant student characteristics (e.g. gender, age, class rank, etc.). Separate regressions are run for proctored and unproctored tests. Harmon and Lambrinos (2008) argue that human capital related variables should have a bigger impact on proctored test scores and, consequently, the explanatory power of the regression model should be higher for proctored than unproctored tests. This is primarily because cheating, which is an unobserved activity/behavior, dilutes the relationship between human capital and test scores in the unproctored setting. They use the R^2 statistic to compare the explanatory power of the regressions with and without proctoring.

For this study, several baseline regressions were estimated that included different combinations of the student characteristics in Tables 1a and 1b However, only a student's human capital (proxied by GPA) and age were found to be significant predictors of performance in various specifications (a student's sex and credit hours earned were statistically insignificant in almost all specifications).[7] Accordingly, the final specifications adopted in this paper are:

$$Score_{up} = a_0 + a_1\ GPA + a_2\ Age + \varepsilon_1 \quad (1)$$

$$Score_{pr} = b_0 + b_1\ GPA + b_2\ Age + \varepsilon_2 \quad (2)$$

where $Score_{up}$is score on an unproctored exam, $Score_{pr}$ is score on a proctored exam and ε_1 and ε_2 are regression errors. Therefore, equations (1) and (2) represent regressions under the unproctored regime and proctored regime, respectively. Since there are three exams, three separate regressions are estimated under each regime, along with a fourth specification that uses the average score of a student in the three exams. Exam scores are measured as a percentage.

After the regressions, we conduct two tests to compare the regression results under (1) and (2). First, following Harmon and Lambrinos (2008) and Arnold (2016), the Goldfeld-Quandt test, originally designed to detect heteroscedasticity, is used to pairwise test equality of error variances in corresponding regressions under the two regimes (with and without proctoring). The premise is that if cheating is prevalent under no proctoring, the error variances in the two models (and, indirectly, the R^2's) would be significantly different.

Second, the Chow test will be used to test the equality of estimated coefficients under the two regimes. Originally designed to test for the presence of a structural break, here the test is used to detect if the impact on performance of human capital (primarily) and age have fundamentally changed (shifted) after the introduction of online proctoring. Accordingly, the test checks if $b_0 = a_0$, $b_1 = a_1$and $b_2 = a_2$ (Wooldridge, 2009).[8] The premise is that if academic dishonesty was prevalent prior to proctoring, online proctoring induces a regime shift – a tighter relationship between human capital and exam performance, and a larger impact of the former on the latter – which would be detected by the Chow test.

4.2.2. Results

The results for economics are presented in Table 3a. As can be seen, GPA is a highly significant predictor of performance in both regimes. In unproctored exams, a unit increase in GPA raises exam score by half a letter grade (5 percentage points) on average. In proctored exams, however, GPA has a bigger impact – close to 8 percentage points, on average. This signifies that there is indeed a stronger relationship between ability and performance when proctoring is in place. Moreover, age also has a positive and significant effect on performance only under proctoring. Overall, it appears that online proctoring reduces the dilution of performance by acts of cheating (or academic dishonesty) and allows traditional variables such as human capital and maturity to have a bigger role as determinants of exam performance.

Further buttressing this assessment, the regression explanatory power (as measured by R^2) is generally higher for proctored exams than unproctored exams (average of 16 percent versus 7 percent across the three exams). To test this formally, a pairwise (between corresponding

[7] In the early stages of the study, preliminary regressions were also run that included several other demographic and academic background variables (e.g. race/ethnicity, first-generation status, transfer status, major, etc.). None of the variables proved to be significant predictors of student performance in the two courses.

[8] In principle, the Chow test is equivalent to introducing a group indicator – in our case a dummy variable that equals one for proctoring, zero otherwise – and its full set of interactions in a pooled model and testing the group significance of the intercept shifter and estimated interaction coefficients. The test produces an F statistic. The results of the pooled model with the full set of interactions are put together in an appendix, available upon request.

Table 3a
Exam score regressions, Economics.

	Unproctored				Proctored			
	Exam 1	Exam 2	Exam 3	Average	Exam 1	Exam 2	Exam 3	Average
GPA	5.672*** (3.51)	5.255*** (3.18)	4.821** (2.42)	5.249*** (3.64)	10.568*** (4.78)	7.250*** (2.57)	5.678** (1.99)	7.832*** (3.78)
Age	−0.030 (0.06)	−0.646 (1.32)	0.170 (0.29)	−0.168 (0.39)	2.338** (2.28)	2.679** (2.05)	2.949** (2.23)	2.655*** (2.77)
Constant	62.835*** (5.69)	80.891*** (7.16)	57.481*** (4.23)	67.069*** (6.80)	−9.741 (0.44)	−10.460 (0.37)	−21.152 (0.75)	−13.784 (0.67)
R^2	0.087	0.088	0.043	0.095	0.265	0.119	0.100	0.216
N	133	133	133	133	75	133	75	75
G-Q test *F(72, 130)*					1.17	1.81***	1.28	1.28
Chow test *F(3,202)*					17.61***	34.22***	30.75***	41.90***

Note: |t| ratios are shown in parentheses. *** shows significance at $p < 0.01$. ** shows significance at $p < 0.05$. The G-Q and Chow tests are for corresponding exams under the proctored and unproctored regimes.

Table 3b
Exam score regressions, Geography.

	Unproctored				Proctored			
	Exam 1	Exam 2	Exam 3	Average	Exam 1	Exam 2	Exam 3	Average
GPA	5.628*** (6.02)	4.757*** (4.78)	3.225*** (3.57)	4.537*** (6.02)	8.856*** (5.68)	7.303*** (4.39)	7.070*** (4.19)	7.743*** (6.00)
Age	0.375** (1.98)	0.473** (2.35)	0.088 (0.48)	0.312** (2.04)	0.092 (0.36)	−0.051 (0.19)	−0.246 (0.89)	−0.069 (0.32)
Constant	57.236*** (12.36)	60.789*** (12.32)	77.664*** (17.35)	65.230*** (17.45)	44.502*** (6.09)	49.859*** (6.39)	58.268*** (7.37)	50.876*** (8.41)
R^2	0.133	0.099	0.046	0.133	0.178	0.116	0.114	0.198
N	288	288	288	288	152	152	152	152
G-Q test *F(149, 285)*					1.39***	1.39***	1.73***	1.46***
Chow test *F(3,434)*					27.25***	56.59***	68.72***	78.25***

Note: |t| ratios are shown in parentheses. *** shows significance at $p < 0.01$. ** shows significance at $p < 0.05$. The G-Q and Chow tests are for corresponding exams under the proctored and unproctored regimes.

exams) test of the equality of error variances is conducted using the Goldfeld-Quandt test. The results, collated in Table 3a, show that statistically the difference is highly significant for Exam 2 but not for the other exams (for Exam 3 and the average score, the F-statistic has a p-value of 0.11, which is just outside of the conventional range). Table 3a also contains results of the Chow test, which checks if the estimated coefficients in the proctored regression, as a group, are statistically different from those in the unproctored regression. As can be seen, the F-statistic is highly statistically significant ($p < 0.05$) for each exam score, confirming a change in coefficients akin to a structural break.

The results for geography are presented in Table 3b. Again, higher GPA is associated with better performance in both the proctored and unproctored environments. But the impact of a unit increase in GPA on exam scores is higher under proctoring than without, and is comparable in magnitude to those recorded for economics (on average about half and three-quarters of a letter grade, respectively, in unproctored and proctored exams). Interestingly, here age is a significant predictor of performance in the unproctored regime, although the nature of its impact is still similar – controlling for ability, older students tend to do better than their younger counterparts.

Given the weaker role of ability in exam performance and the generally lower explanatory power of the unproctored regressions, one can again surmise that cheating was likely taking place in the geography course prior to the introduction of online proctoring. The two formal tests

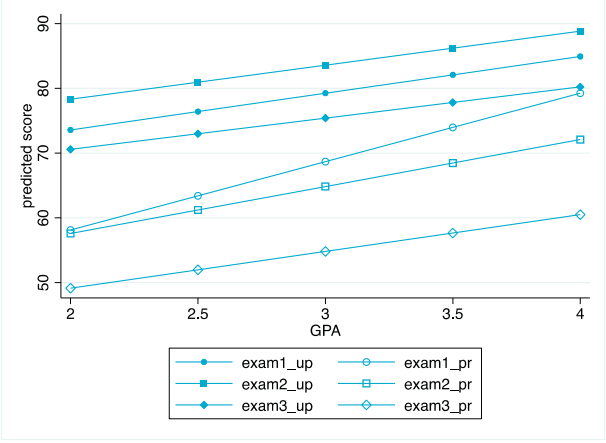


Fig. 2a. GPA and exam performance regression fits, Economics.

reinforce this assessment. The Goldfeld-Quandt test of error variances resulted in F-statistics that were significant at the 1% level in all three exams (and for average score). With similarly highly significant statistics, the Chow test also indicated that estimated coefficients under the two regimes were, as a group, unequal.[9]

Finally, to help visualize these results, Fig. 2a–b plots the predicted GPA-exam score relationship for each exam by proctoring status. Fig. 2a shows the fitted lines for economics (based on the estimates in Table 3a) and Fig. 2b shows those for geography (based on the estimates in Table 3b). As expected, two observations stand out in these figures. First, the fitted lines for unproctored exams lie well above those for proctored

[9] The large F-statistics of the Chow test were not merely a result of a significant intercept shifter (the dummy for online proctoring). In geography, for instance, the interaction of GPA and online proctoring was statistically significant – hence the impact of GPA statistically higher – in two of the 3 Em as well as in the average exam score regression.

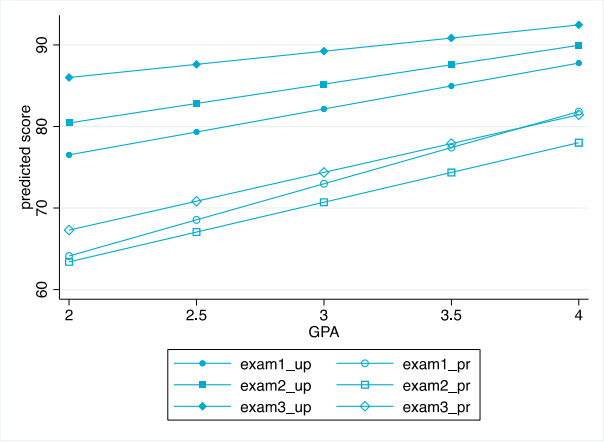


Fig. 2b. GPA and exam performance regression fits, Geography.

exams, revealing the gap in average predicted score at every level of GPA between the two regimes. Second, as evidenced by the steeper slopes of the fitted lines for proctored exams, the GPA-exam score relationship is tighter (more significant) with proctoring than without.

5. Caveats and limitations

A few caveats and limitations of the study are in order and point to future research questions. First, it is possible that some of the drop in performance under proctoring can be due to the added stress or anxiety that students could experience when being recorded during an exam (Woldeab & Brothen, 2019; Butler-Henderson & Crawford, 2020). Given the high comfort level that today's generation of students have with this type of technology, however, this effect may be minimal. Moreover, the instructors did not receive any feedback indicating that a majority of students were discomforted by the experience.

Second, the analyses, results and implications of the study are constrained by the type of assessments that were employed in the two courses, which were mostly comprised of objective (multiple choice) questions. It can be argued that these types of assessments are more amenable to cheating and, therefore, the effect of proctoring is estimated where it is precisely most useful. In other words, estimates of the extent of cheating in online courses documented in this paper are perhaps upper-bound estimates. It would be interesting to measure the amount of cheating (and hence the value of proctoring) in other types of assessments.

Third, whereas the study design in this paper has, as noted above, many desirable attributes, it also has certain limitations. For instance, it is still possible, although unlikely, that the student composition before and after proctoring differed in systematic ways that we could not account for with the covariates in our regressions. Also, based on this study one cannot directly infer the efficacy of online proctoring vis-à-vis live (F2F) supervision. In general, to make broader and far-ranging inferences, a study ideally ought to be truly experimental, assigning students randomly into a control group (unmonitored online) and two types of treatments (online and F2F monitoring), and should evaluate performance using a difference-in-difference (DiD) type analysis. It would also be interesting to study the efficacy of web-based proctoring longitudinally (over a longer period of time), as students adapt to the system and potentially adjust their behavior.

6. Conclusions and implications

As online education expands apace in all types of institutions of higher learning, faculty and administrators continue to grapple with various issues surrounding the medium, including quality of delivery and learning, engagement, equity, accessibility and integrity. The aim of these efforts, ultimately, is to continuously improve all aspects of distance education so that the main stakeholders – students, faculty and institutions – are comfortable in their experiences and satisfied in the quality of teaching and learning that takes place.

This paper has presented new evidence on one aspect of distance education that will inform this continuous improvement process, which is academic integrity in online courses. Although the early literature was somewhat inconclusive on the extent of cheating in online courses, recently there is mounting evidence that it is substantial (Sullivan, 2016; Norris, 2019; Golden & Kohlbeck, 2020). Relative to the extant literature, the evidence in this paper adds value in several ways. The study is based on two courses that come from different schools (colleges) in a medium size comprehensive university, ensuring that results are unlikely to be driven by some subject- or discipline-specific phenomenon. It also uses data encompassing several semesters, which means the findings are temporally robust and not a snapshot reflection of a given semester or period.

To gauge cheating, the study took advantage of a quasiexperiment in which online proctoring was introduced at a particular point in the course of the study period. This enabled comparison of average performance pre- and post-proctoring, given the absence of any commensurate change in course structure, content or format. We also account for basic student demographics and academic background, which indicated that there were no substantive changes to sample composition that happened concurrently or because of the advent of proctoring, implying the analysis was unlikely to be affected by sample selection. The fact that proctoring was not associated with a change in medium also addressed a pointed critique in the related literature, where studies typically compared performance in unproctored online assessments to that in proctored F2F assessments, making the disentangling of the effects of proctoring and exam environment difficult (Fask et al., 2014). Such studies also faced a more acute sample selection problem.

Given these features, the results of the analyses strongly suggest that cheating was taking place prior to the introduction of online proctoring. The average score under proctoring was significantly lower in every exam that was given in the two courses. Regression exercises also showed that GPA, as a measure of ability, had a larger effect on proctored exam scores than unproctored ones. Relevant statistical tests mostly confirmed this, showing a structural break in coefficient estimates and higher overall explanatory power of regressions involving proctored exam scores.

The main implication of these results is that academic dishonesty is indeed a serious issue in online courses. Despite a series of mitigation measures that were adopted without direct proctoring – such as the use of a special browser, a restricted testing period, randomized questions and choices, and a strict timer – it appears that cheating was relatively commonplace. Cheating apparently also paid off handsomely, at least when it comes to exam performance, often raising scores by about a letter grade.[10] A related implication is that some form of direct proctoring is perhaps the most effective way of mitigating cheating during high-stakes online assessments. Other tangential measures such as those listed above can be helpful but should be thought of as a complement, not a replacement, to direct proctoring.

Another important implication is that for proctoring to be an effective hindrance to academic dishonesty, it should not necessarily take the form

[10] The results and main implication of this paper are in line with the findings of a recent meta-analysis of several studies that examined test scores in proctored and unproctored assessments (Steger et al., 2020). The underlying studies mostly involved cognitive, reasoning and other ability tests, although there were also subject-based assessments. The meta-analysis documented a pooled effect of 20%, showing higher scores in unproctored tests. The result demonstrated that unproctored online ability tests are generally biased. Interestingly, the bias is significantly smaller for assessment measures that are difficult to research on the internet.

of in-person or live supervision. Typically proctoring is thought to involve supervision of an assessment by an instructor or an assignee in a F2F setting (on campus or at a testing center) or through remote monitoring. But this type of proctoring can be infeasible (e.g. for fully online students) and is also quite expensive (Lee-Post & Hapke, 2017; Norris, 2019). The inconvenience and costs associated with live proctoring can therefore play a role in discouraging demand for online education by students. They also likely discourage instructors and institutions who would otherwise have ventured into or expanded distance education except for their skepticism of the integrity of online courses without proctored assessments.

The fact that a technological solution such as the one examined in this study (online proctoring through a webcam recording software) does an effective job in mitigating academic dishonesty is thus reassuring for all stakeholders. The results in this paper do not suggest that the solution is perfect – for that matter, there is no evidence that in-person monitoring is either – but they are significant enough to indicate its efficacy. Coupled with the relatively low-cost, user friendly nature of this type of technology, the results should broadly encourage its adoption by concerned faculty and institutions.

Declaration of competing interest

The authors have no conflict of interest.

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